

The van Kampen Theorem

Suppose a space X is decomposed as the union of a collection of path-connected open subsets A_α , each of which contains the basepoint $x_0 \in X$. By the remarks in the preceding paragraph, the homomorphisms $j_\alpha: \pi_1(A_\alpha) \rightarrow \pi_1(X)$ induced by the inclusions $A_\alpha \hookrightarrow X$ extend to a homomorphism $\Phi: *_{\alpha} \pi_1(A_\alpha) \rightarrow \pi_1(X)$. The van Kampen theorem will say that Φ is very often surjective, but we can expect Φ to have a nontrivial kernel in general. For if $i_{\alpha\beta}: \pi_1(A_\alpha \cap A_\beta) \rightarrow \pi_1(A_\alpha)$ is the homomorphism induced by the inclusion $A_\alpha \cap A_\beta \hookrightarrow A_\alpha$ then $j_\alpha i_{\alpha\beta} = j_\beta i_{\beta\alpha}$, both these compositions being induced by the inclusion $A_\alpha \cap A_\beta \hookrightarrow X$, so the kernel of Φ contains all the elements of the form $i_{\alpha\beta}(\omega) i_{\beta\alpha}(\omega)^{-1}$ for $\omega \in \pi_1(A_\alpha \cap A_\beta)$. Van Kampen's theorem asserts that under fairly broad hypotheses this gives a full description of Φ :

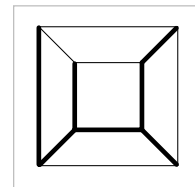
Theorem 1.20. *If X is the union of path-connected open sets A_α each containing the basepoint $x_0 \in X$ and if each intersection $A_\alpha \cap A_\beta$ is path-connected, then the homomorphism $\Phi: *_{\alpha} \pi_1(A_\alpha) \rightarrow \pi_1(X)$ is surjective. If in addition each intersection $A_\alpha \cap A_\beta \cap A_\gamma$ is path-connected, then the kernel of Φ is the normal subgroup N generated by all elements of the form $i_{\alpha\beta}(\omega) i_{\beta\alpha}(\omega)^{-1}$, and so Φ induces an isomorphism $\pi_1(X) \approx *_{\alpha} \pi_1(A_\alpha) / N$.*

Example 1.21: Wedge Sums. In Chapter 0 we defined the wedge sum $\bigvee_{\alpha} X_{\alpha}$ of a collection of spaces X_{α} with basepoints $x_{\alpha} \in X_{\alpha}$ to be the quotient space of the disjoint union $\bigsqcup_{\alpha} X_{\alpha}$ in which all the basepoints x_{α} are identified to a single point. If each x_{α} is a deformation retract of an open neighborhood U_{α} in X_{α} , then X_{α} is a deformation retract of its open neighborhood $A_{\alpha} = X_{\alpha} \bigvee_{\beta \neq \alpha} U_{\beta}$. The intersection of two or more distinct A_{α} 's is $\bigvee_{\alpha} U_{\alpha}$, which deformation retracts to a point. Van Kampen's theorem then implies that $\Phi: *_{\alpha} \pi_1(X_{\alpha}) \rightarrow \pi_1(\bigvee_{\alpha} X_{\alpha})$ is an isomorphism.

Thus for a wedge sum $\bigvee_{\alpha} S_{\alpha}^1$ of circles, $\pi_1(\bigvee_{\alpha} S_{\alpha}^1)$ is a free group, the free product of copies of $\mathbb{Z}$, one for each circle S_{α}^1 . In particular, $\pi_1(S^1 \vee S^1)$ is the free group $\mathbb{Z} * \mathbb{Z}$, as in the example at the beginning of this section.

It is true more generally that the fundamental group of any connected graph is free, as we show in §1.A. Here is an example illustrating the general technique.

Example 1.22. Let X be the graph shown in the figure, consisting of the twelve edges of a cube. The seven heavily shaded edges form a maximal tree $T \subset X$, a contractible subgraph containing all the vertices of X . We claim that $\pi_1(X)$ is the free product of five

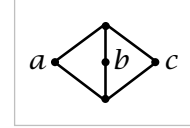


copies of $\mathbb{Z}$, one for each edge not in T . To deduce this from van Kampen's theorem, choose for each edge e_{α} of $X - T$ an open neighborhood A_{α} of $T \cup e_{\alpha}$ in X that deformation retracts onto $T \cup e_{\alpha}$. The intersection of two or more A_{α} 's deformation retracts onto T , hence is contractible. The A_{α} 's form a cover of X satisfying the hypotheses of van Kampen's theorem, and since the intersection of

any two of them is simply-connected we obtain an isomorphism $\pi_1(X) \approx *_\alpha \pi_1(A_\alpha)$. Each A_α deformation retracts onto a circle, so $\pi_1(X)$ is free on five generators, as claimed. As explicit generators we can choose for each edge e_α of $X - T$ a loop f_α that starts at a basepoint in T , travels in T to one end of e_α , then across e_α , then back to the basepoint along a path in T .

Van Kampen's theorem is often applied when there are just two sets A_α and A_β in the cover of X , so the condition on triple intersections $A_\alpha \cap A_\beta \cap A_\gamma$ is superfluous and one obtains an isomorphism $\pi_1(X) \approx (\pi_1(A_\alpha) * \pi_1(A_\beta))/N$, under the assumption that $A_\alpha \cap A_\beta$ is path-connected. The proof in this special case is virtually identical with the proof in the general case, however.

One can see that the intersections $A_\alpha \cap A_\beta$ need to be path-connected by considering the example of S^1 decomposed as the union of two open arcs. In this case Φ is not surjective. For an example showing that triple intersections $A_\alpha \cap A_\beta \cap A_\gamma$ need to be path-connected, let X be the suspension of three points a, b, c , and let A_α, A_β , and A_γ be the complements of these three points. The theorem does apply to the covering $\{A_\alpha, A_\beta\}$, so there are isomorphisms $\pi_1(X) \approx \pi_1(A_\alpha) * \pi_1(A_\beta) \approx \mathbb{Z} * \mathbb{Z}$ since $A_\alpha \cap A_\beta$ is contractible. If we tried to use the covering $\{A_\alpha, A_\beta, A_\gamma\}$, which has each of the twofold intersections path-connected but not the triple intersection, then we would get $\pi_1(X) \approx \mathbb{Z} * \mathbb{Z} * \mathbb{Z}$, but this is not isomorphic to $\mathbb{Z} * \mathbb{Z}$ since it has a different abelianization.

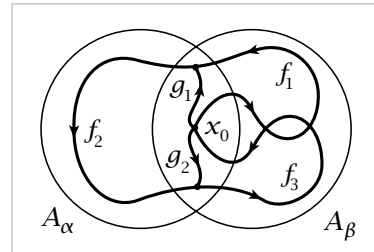


Proof of van Kampen's theorem: First we consider surjectivity of Φ . Given a loop $f: I \rightarrow X$ at the basepoint x_0 , we claim there is a partition $0 = s_0 < s_1 < \dots < s_m = 1$ of I such that each subinterval $[s_{i-1}, s_i]$ is mapped by f to a single A_α . Namely, since f is continuous, each $s \in I$ has an open neighborhood V_s in I mapped by f to some A_α . We may in fact take V_s to be an interval whose closure is mapped to a single A_α . Compactness of I implies that a finite number of these intervals cover I . The endpoints of this finite set of intervals then define the desired partition of I .

Denote the A_α containing $f([s_{i-1}, s_i])$ by A_i , and let f_i be the path obtained by restricting f to $[s_{i-1}, s_i]$. Then f is the composition $f_1 \cdot \dots \cdot f_m$ with f_i a path in A_i . Since we assume $A_i \cap A_{i+1}$ is path-connected, we may choose a path g_i in $A_i \cap A_{i+1}$ from x_0 to the point $f(s_i) \in A_i \cap A_{i+1}$. Consider the loop

$$(f_1 \cdot \bar{g}_1) \cdot (g_1 \cdot f_2 \cdot \bar{g}_2) \cdot (g_2 \cdot f_3 \cdot \bar{g}_3) \cdot \dots \cdot (g_{m-1} \cdot f_m)$$

which is homotopic to f . This loop is a composition of loops each lying in a single A_i , the loops indicated by the parentheses. Hence $[f]$ is in the image of Φ , and Φ is surjective.



The harder part of the proof is to show that the kernel of Φ is N . It may clarify

matters to introduce some terminology. By a *factorization* of an element $[f] \in \pi_1(X)$ we shall mean a formal product $[f_1] \cdots [f_k]$ where:

- Each f_i is a loop in some A_α at the basepoint x_0 , and $[f_i] \in \pi_1(A_\alpha)$ is the homotopy class of f_i .
- The loop f is homotopic to $f_1 \cdots f_k$ in X .

A factorization of $[f]$ is thus a word in $*_\alpha \pi_1(A_\alpha)$, possibly unreduced, that is mapped to $[f]$ by Φ . The proof of surjectivity of Φ showed that every $[f] \in \pi_1(X)$ has a factorization.

We will be concerned now with the uniqueness of factorizations. Call two factorizations of $[f]$ *equivalent* if they are related by a sequence of the following two sorts of moves or their inverses:

- Combine adjacent terms $[f_i][f_{i+1}]$ into a single term $[f_i \cdot f_{i+1}]$ if $[f_i]$ and $[f_{i+1}]$ lie in the same group $\pi_1(A_\alpha)$.
- Regard the term $[f_i] \in \pi_1(A_\alpha)$ as lying in the group $\pi_1(A_\beta)$ rather than $\pi_1(A_\alpha)$ if f_i is a loop in $A_\alpha \cap A_\beta$.

The first move does not change the element of $*_\alpha \pi_1(A_\alpha)$ defined by the factorization. The second move does not change the image of this element in the quotient group $Q = *_\alpha \pi_1(A_\alpha)/N$, by the definition of N . So equivalent factorizations give the same element of Q .

If we can show that any two factorizations of $[f]$ are equivalent, this will say that the map $Q \rightarrow \pi_1(X)$ induced by Φ is injective, hence the kernel of Φ is exactly N , and the proof will be complete.

Let $[f_1] \cdots [f_k]$ and $[f'_1] \cdots [f'_\ell]$ be two factorizations of $[f]$. The composed paths $f_1 \cdots f_k$ and $f'_1 \cdots f'_\ell$ are then homotopic, so let $F: I \times I \rightarrow X$ be a homotopy from $f_1 \cdots f_k$ to $f'_1 \cdots f'_\ell$. There exist partitions $0 = s_0 < s_1 < \cdots < s_m = 1$ and $0 = t_0 < t_1 < \cdots < t_n = 1$ such that each rectangle $R_{ij} = [s_{i-1}, s_i] \times [t_{j-1}, t_j]$ is mapped by F into a single A_α , which we label A_{ij} . These partitions may be obtained by covering $I \times I$ by finitely many rectangles $[a, b] \times [c, d]$ each mapping to a single A_α , using a compactness argument, then partitioning $I \times I$ by the union of all the horizontal and vertical lines containing edges of these rectangles. We may assume the s -partition subdivides the partitions giving the products $f_1 \cdots f_k$ and $f'_1 \cdots f'_\ell$. Since F maps a neighborhood of R_{ij} to A_{ij} , we may perturb the vertical sides of the rectangles R_{ij} so that each point of $I \times I$ lies in at most three R_{ij} 's. We may assume there are at least three rows of rectangles, so we can do this perturbation just on the rectangles in the intermediate rows, leaving the top and bottom rows unchanged. Let us relabel the new rectangles $R_1, R_2, \dots, R_{mn}$, ordering them as in the figure.

9	10	11	12
5	6	7	8
1	2	3	4

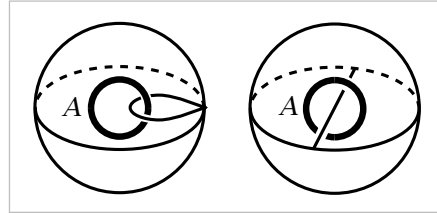
If γ is a path in $I \times I$ from the left edge to the right edge, then the restriction $F|_{\gamma}$ is a loop at the basepoint x_0 since F maps both the left and right edges of $I \times I$ to x_0 . Let γ_r be the path separating the first r rectangles $R_1, \dots, R_r$ from the remaining rectangles. Thus γ_0 is the bottom edge of $I \times I$ and γ_{mn} is the top edge. We pass from γ_r to γ_{r+1} by pushing across the rectangle R_{r+1} .

Let us call the corners of the R_r 's *vertices*. For each vertex v with $F(v) \neq x_0$, let g_v be a path from x_0 to $F(v)$. We can choose g_v to lie in the intersection of the two or three A_{ij} 's corresponding to the R_r 's containing v since we assume the intersection of any two or three A_{ij} 's is path-connected. If we insert into $F|_{\gamma_r}$ the appropriate paths $\overline{g}_v g_v$ at successive vertices, as in the proof of surjectivity of Φ , then we obtain a factorization of $[F|_{\gamma_r}]$ by regarding the loop corresponding to a horizontal or vertical segment between adjacent vertices as lying in the A_{ij} for either of the R_s 's containing the segment. Different choices of these containing R_s 's change the factorization of $[F|_{\gamma_r}]$ to an equivalent factorization. Furthermore, the factorizations associated to successive paths γ_r and γ_{r+1} are equivalent since pushing γ_r across R_{r+1} to γ_{r+1} changes $F|_{\gamma_r}$ to $F|_{\gamma_{r+1}}$ by a homotopy within the A_{ij} corresponding to R_{r+1} , and we can choose this A_{ij} for all the segments of γ_r and γ_{r+1} in R_{r+1} .

We can arrange that the factorization associated to γ_0 is equivalent to the factorization $[f_1] \cdots [f_k]$ by choosing the path g_v for each vertex v along the lower edge of $I \times I$ to lie not just in the two A_{ij} 's corresponding to the R_s 's containing v , but also to lie in the A_α for the f_i containing v in its domain. In case v is the common end-point of the domains of two consecutive f_i 's we have $F(v) = x_0$, so there is no need to choose a g_v . In similar fashion we may assume that the factorization associated to the final γ_{mn} is equivalent to $[f'_1] \cdots [f'_\ell]$. Since the factorizations associated to all the γ_r 's are equivalent, we conclude that the factorizations $[f_1] \cdots [f_k]$ and $[f'_1] \cdots [f'_\ell]$ are equivalent. $\square$

Example 1.23: Linking of Circles. We can apply van Kampen's theorem to calculate the fundamental groups of three spaces discussed in the introduction to this chapter, the complements in $\mathbb{R}^3$ of a single circle, two unlinked circles, and two linked circles.

The complement $\mathbb{R}^3 - A$ of a single circle A deformation retracts onto a wedge sum $S^1 \vee S^2$ embedded in $\mathbb{R}^3 - A$ as shown in the first of the two figures at the right. It may be easier to see that $\mathbb{R}^3 - A$ deformation retracts onto the union of S^2 with a diameter, as in the second figure,

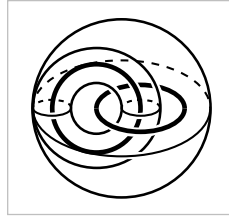


where points outside S^2 deformation retract onto S^2 , and points inside S^2 and not in A can be pushed away from A toward S^2 or the diameter. Having this deformation retraction in mind, one can then see how it must be modified if the two endpoints of the diameter are gradually moved toward each other along the equator until they coincide, forming the S^1 summand of $S^1 \vee S^2$. Another way of seeing the deformation

retraction of $\mathbb{R}^3 - A$ onto $S^1 \vee S^2$ is to note first that an open ε -neighborhood of $S^1 \vee S^2$ obviously deformation retracts onto $S^1 \vee S^2$ if ε is sufficiently small. Then observe that this neighborhood is homeomorphic to $\mathbb{R}^3 - A$ by a homeomorphism that is the identity on $S^1 \vee S^2$. In fact, the neighborhood can be gradually enlarged by homeomorphisms until it becomes all of $\mathbb{R}^3 - A$.

In any event, once we see that $\mathbb{R}^3 - A$ deformation retracts to $S^1 \vee S^2$, then we immediately obtain isomorphisms $\pi_1(\mathbb{R}^3 - A) \approx \pi_1(S^1 \vee S^2) \approx \mathbb{Z}$ since $\pi_1(S^2) = 0$.

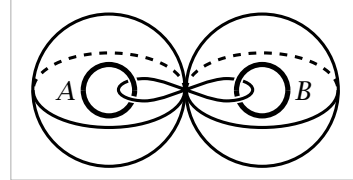
In similar fashion, the complement $\mathbb{R}^3 - (A \cup B)$ of two unlinked circles A and B deformation retracts onto $S^1 \vee S^1 \vee S^2 \vee S^2$, as in the figure to the right. From



this we get $\pi_1(\mathbb{R}^3 - (A \cup B)) \approx$

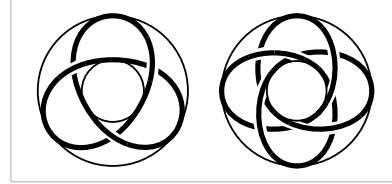
$\mathbb{Z} * \mathbb{Z}$. On the other hand, if A

and B are linked, then $\mathbb{R}^3 - (A \cup B)$ deformation retracts onto the wedge sum of S^2 and a torus $S^1 \times S^1$ separating A and B , as shown in the figure to the left, hence $\pi_1(\mathbb{R}^3 - (A \cup B)) \approx \pi_1(S^1 \times S^1) \approx \mathbb{Z} \times \mathbb{Z}$.



Example 1.24: Torus Knots. For relatively prime positive integers m and n , the **torus knot** $K = K_{m,n} \subset \mathbb{R}^3$ is the image of the embedding $f: S^1 \rightarrow S^1 \times S^1 \subset \mathbb{R}^3$, $f(z) = (z^m, z^n)$, where the torus $S^1 \times S^1$ is embedded in $\mathbb{R}^3$ in the standard way.

The knot K winds around the torus a total of m times in the longitudinal direction and n times in the meridional direction, as shown in the figure for the cases $(m, n) = (2, 3)$ and $(3, 4)$. One needs to assume that m and n are relatively prime in order



for the map f to be injective. Without this assumption f would be d -to-1 where d is the greatest common divisor of m and n , and the image of f would be the knot $K_{m/d, n/d}$. One could also allow negative values for m or n , but this would only change K to a mirror-image knot.

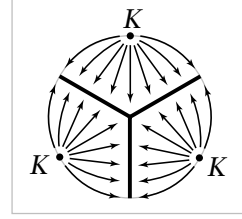
Let us compute $\pi_1(\mathbb{R}^3 - K)$. It is slightly easier to do the calculation with $\mathbb{R}^3$ replaced by its one-point compactification S^3 . An application of van Kampen's theorem shows that this does not affect π_1 . Namely, write $S^3 - K$ as the union of $\mathbb{R}^3 - K$ and an open ball B formed by the compactification point together with the complement of a large closed ball in $\mathbb{R}^3$ containing K . Both B and $B \cap (\mathbb{R}^3 - K)$ are simply-connected, the latter space being homeomorphic to $S^2 \times \mathbb{R}$. Hence van Kampen's theorem implies that the inclusion $\mathbb{R}^3 - K \hookrightarrow S^3 - K$ induces an isomorphism on π_1 .

We compute $\pi_1(S^3 - K)$ by showing that it deformation retracts onto a 2-dimensional complex $X = X_{m,n}$ homeomorphic to the quotient space of a cylinder $S^1 \times I$ under the identifications $(z, 0) \sim (e^{2\pi i/m} z, 0)$ and $(z, 1) \sim (e^{2\pi i/n} z, 1)$. If we let X_m and X_n be the two halves of X formed by the quotients of $S^1 \times [0, 1/2]$ and $S^1 \times [1/2, 1]$, then X_m and X_n are the mapping cylinders of $z \mapsto z^m$ and $z \mapsto z^n$. The intersection

$X_m \cap X_n$ is the circle $S^1 \times \{1/2\}$, the domain end of each mapping cylinder.

To obtain an embedding of X in $S^3 - K$ as a deformation retract we will use the standard decomposition of S^3 into two solid tori $S^1 \times D^2$ and $D^2 \times S^1$, the result of regarding S^3 as $\partial D^4 = \partial(D^2 \times D^2) = \partial D^2 \times D^2 \cup D^2 \times \partial D^2$. Geometrically, the first solid torus $S^1 \times D^2$ can be identified with the compact region in $\mathbb{R}^3$ bounded by the standard torus $S^1 \times S^1$ containing K , and the second solid torus $D^2 \times S^1$ is then the closure of the complement of the first solid torus, together with the compactification point at infinity. Notice that meridional circles in $S^1 \times S^1$ bound disks in the first solid torus, while it is longitudinal circles that bound disks in the second solid torus.

In the first solid torus, K intersects each of the meridian circles $\{x\} \times \partial D^2$ in m equally spaced points, as indicated in the figure at the right, which shows a meridian disk $\{x\} \times D^2$. These m points can be separated by a union of m radial line segments. Letting x vary, these radial segments then trace out a copy of the mapping cylinder X_m in the first solid torus. Symmetrically, there is a copy of the other mapping cylinder X_n in the second solid torus. The complement of K in the first solid torus deformation retracts onto X_m by flowing within each meridian disk as shown. In similar fashion the complement of K in the second solid torus deformation retracts onto X_n . These two deformation retractions do not agree on their common domain of definition $S^1 \times S^1 - K$, but this is easy to correct by distorting the flows in the two solid tori so that in $S^1 \times S^1 - K$ both flows are orthogonal to K . After this modification we now have a well-defined deformation retraction of $S^3 - K$ onto X . Another way of describing the situation would be to say that for an open ε -neighborhood N of K bounded by a torus T , the complement $S^3 - N$ is the mapping cylinder of a map $T \rightarrow X$.



To compute $\pi_1(X)$ we apply van Kampen's theorem to the decomposition of X as the union of X_m and X_n , or more properly, open neighborhoods of these two sets that deformation retract onto them. Both X_m and X_n are mapping cylinders that deformation retract onto circles, and $X_m \cap X_n$ is a circle, so all three of these spaces have fundamental group $\mathbb{Z}$. A loop in $X_m \cap X_n$ representing a generator of $\pi_1(X_m \cap X_n)$ is homotopic in X_m to a loop representing m times a generator, and in X_n to a loop representing n times a generator. Van Kampen's theorem then says that $\pi_1(X)$ is the quotient of the free group on generators a and b obtained by factoring out the normal subgroup generated by the element $a^m b^{-n}$.

Let us denote by $G_{m,n}$ this group $\pi_1(X_{m,n})$ defined by two generators a and b and one relation $a^m = b^n$. If m or n is 1, then $G_{m,n}$ is infinite cyclic since in these cases the relation just expresses one generator as a power of the other. To describe the structure of $G_{m,n}$ when $m, n > 1$ let us first compute the center of $G_{m,n}$, the subgroup consisting of elements that commute with all elements of $G_{m,n}$. The element $a^m = b^n$ commutes with a and b , so the cyclic subgroup C generated

by this element lies in the center. In particular, C is a normal subgroup, so we can pass to the quotient group $G_{m,n}/C$, which is the free product $\mathbb{Z}_m * \mathbb{Z}_n$. According to Exercise 1 at the end of this section, a free product of nontrivial groups has trivial center. From this it follows that C is exactly the center of $G_{m,n}$. As we will see in Example 1.44, the elements a and b have infinite order in $G_{m,n}$, so C is infinite cyclic, but we will not need this fact here.

We will show now that the integers m and n are uniquely determined by the group $\mathbb{Z}_m * \mathbb{Z}_n$, hence also by $G_{m,n}$. The abelianization of $\mathbb{Z}_m * \mathbb{Z}_n$ is $\mathbb{Z}_m \times \mathbb{Z}_n$, of order mn , so the product mn is uniquely determined by $\mathbb{Z}_m * \mathbb{Z}_n$. To determine m and n individually, we use another assertion from Exercise 1 at the end of the section, that all torsion elements of $\mathbb{Z}_m * \mathbb{Z}_n$ are conjugate to elements of the subgroups $\mathbb{Z}_m$ and $\mathbb{Z}_n$, hence have order dividing m or n . Thus the maximum order of torsion elements of $\mathbb{Z}_m * \mathbb{Z}_n$ is the larger of m and n . The larger of these two numbers is therefore uniquely determined by the group $\mathbb{Z}_m * \mathbb{Z}_n$, hence also the smaller since the product is uniquely determined.

The preceding analysis of $\pi_1(X_{m,n})$ did not need the assumption that m and n are relatively prime, which was used only to relate $X_{m,n}$ to torus knots. An interesting fact is that $X_{m,n}$ can be embedded in $\mathbb{R}^3$ only when m and n are relatively prime. This is shown in the remarks following Corollary 3.45. For example, $X_{2,2}$ is the Klein bottle since it is the union of two copies of the Möbius band X_2 with their boundary circles identified, so this nonembeddability statement generalizes the fact that the Klein bottle cannot be embedded in $\mathbb{R}^3$.

An algorithm for computing a presentation for $\pi_1(\mathbb{R}^3 - K)$ for an arbitrary smooth or piecewise linear knot K is described in the exercises, but the problem of determining when two of these fundamental groups are isomorphic is generally much more difficult than in the special case of torus knots.

Example 1.25: The Shrinking Wedge of Circles. Consider the subspace $X \subset \mathbb{R}^2$ that is the union of the circles C_n of radius $1/n$ and center $(1/n, 0)$ for $n = 1, 2, \dots$. At first glance one might confuse X with the wedge sum of an infinite sequence of circles, but we will show that X has a much larger fundamental group than the wedge sum. Consider the retractions $r_n: X \rightarrow C_n$ collapsing all C_i 's except C_n to the origin. Each r_n induces a surjection $\rho_n: \pi_1(X) \rightarrow \pi_1(C_n) \approx \mathbb{Z}$, where we take the origin as the basepoint. The product of the ρ_n 's is a homomorphism $\rho: \pi_1(X) \rightarrow \prod_{\infty} \mathbb{Z}$ to the direct product (not the direct sum) of infinitely many copies of $\mathbb{Z}$, and ρ is surjective since for every sequence of integers k_n we can construct a loop $f: I \rightarrow X$ that wraps k_n times around C_n in the time interval $[1 - 1/n, 1 - 1/(n+1)]$. This infinite composition of loops is certainly continuous at each time less than 1, and it is continuous at time 1 since every neighborhood of the basepoint in X contains all but finitely many of the circles C_n . Since $\pi_1(X)$ maps onto the uncountable group $\prod_{\infty} \mathbb{Z}$, it is uncountable.

